

EC-467 Mobile Application Development



BS (CE)-2021 LAB REPORT #7

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

Lab Instructor.

Dr. Muhammad Bilal

LAB REPORT #7

Shared Preferences

Introduction:

Android provides many ways of storing data of an application. One of this way is called Shared Preferences. Shared Preferences allow you to save and retrieve data in the form of key, value pair. The first parameter is the key and the second parameter is the MODE. You can save something in the sharedpreferences by using SharedPreferences. Editor class. We will call the edit method of SharedPreferences instance and will receive it in an editor object.

Its syntax is `Editor editor = sharedPreferences.edit(); editor.putString("key", "value"); editor.commit();`

Objectives:

- To get basic understanding regarding Shared Preferences
- To get basic understanding regarding how data is saved and retrieved
- To get to know about the benefits related to Shared Preferences.

Software used:

Android Studio Iguana

Class Task:

Shared Preferences:

XML CODE OF MAIN ACTIVITY:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    tools:context=".MainActivity">
```

```
<EditText
    android:id="@+id/user_name"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:ems="10"
    android:inputType="textPersonName"
    android:text="User Name" />
<EditText
    android:id="@+id/pass_word"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:ems="10"
    android:inputType="textPersonName"
    android:text="Pass Word" />
<Button
    android:id="@+id/save_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:onClick="saveButton"
    android:text="Save" />
<Button
    android:id="@+id/restore_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:onClick="restoreButton"
    android:text="Get" />
<Button
    android:id="@+id/delete_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:onClick="deleteButton"
    android:text="Delete" />
</LinearLayout>
```

JAVA CODE OF MAIN ACTIVITY:

```
package com.huzaifashahbaz.SharedPreferences;
import android.os.Bundle;
```

```

import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
class PrefActivity extends AppCompatActivity {
    EditText userName, password;
    Button save, restore, delete;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        userName = findViewById(R.id.user_name);
        password = findViewById(R.id.pass_word);
        save = findViewById(R.id.save_btn);
        restore = findViewById(R.id.restore_btn);
        delete = findViewById(R.id.delete_btn);
    }
    public void saveButton(View view){
        String userNameStr = userName.getText().toString();
        String passWordStr = password.getText().toString();
        if(!userNameStr.isEmpty() &&
            !passWordStr.isEmpty()){
            if(MySharedPrefs.setUserNamePW(this, userNameStr, passWordStr)){
                Toast.makeText(this, "Successfully Saved",
                    Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "Not Successfully Saved",
                    Toast.LENGTH_SHORT).show();
            }
        }
    }
    public void restoreButton(View view){
        String[] userNamePW = MySharedPrefs.getUserNamePW(this);
        if(!userNamePW[0].isEmpty() && !userNamePW[1].isEmpty()){
            userName.setText(userNamePW[0]);
            password.setText(userNamePW[1]);
        }
    }
    public void deleteButton(View view){

```

```

        if(MySharedPrefs.removeUserNamePW(this)){
            userName.setText("");
            password.setText("");
            Toast.makeText(this, "Successfully Deleted",
                Toast.LENGTH_SHORT).show();
        } else {
            Toast.makeText(this, "Not Successfully Deleted",
                Toast.LENGTH_SHORT).show();
        }
    }
}

```

JAVA CODE OF MY Shared Preferences CLASS:

```

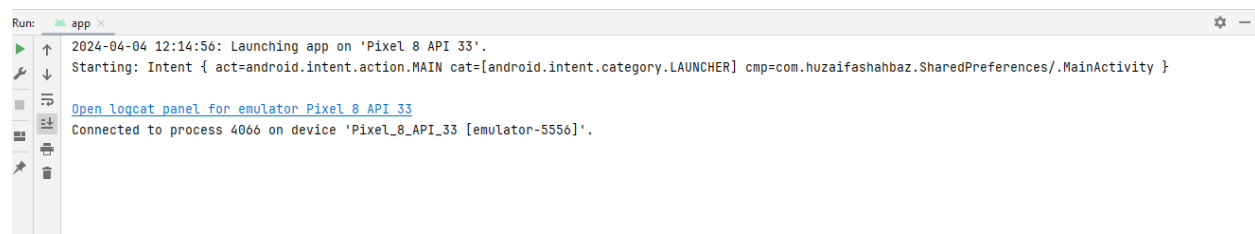
package com.huzaifashahbaz.SharedPreferences;
import android.content.Context;
import android.content.SharedPreferences;
public class MySharedPrefs {
    public static final String USER_NAME = "userName";
    public static final String PASS_WORD = "passWord";
    public static boolean setUserNamePW(Context context, String userName, String
        passWord){
        SharedPreferences prefs = context.getSharedPreferences("MyPrefs",
            context.MODE_PRIVATE);
        SharedPreferences.Editor editor = prefs.edit();

        editor.putString(USER_NAME, userName);
        editor.putString(PASS_WORD, passWord);
        return editor.commit();
    }
    public static String[] getUserNamePW(Context context){
        SharedPreferences prefs = context.getSharedPreferences("MyPrefs",
            context.MODE_PRIVATE);
        String[] userNamePW = new String[2];
        userNamePW[0] = prefs.getString(USER_NAME, "");
        userNamePW[1] = prefs.getString(PASS_WORD, "");
        return userNamePW;
    }
    public static boolean removeUserNamePW(Context context){

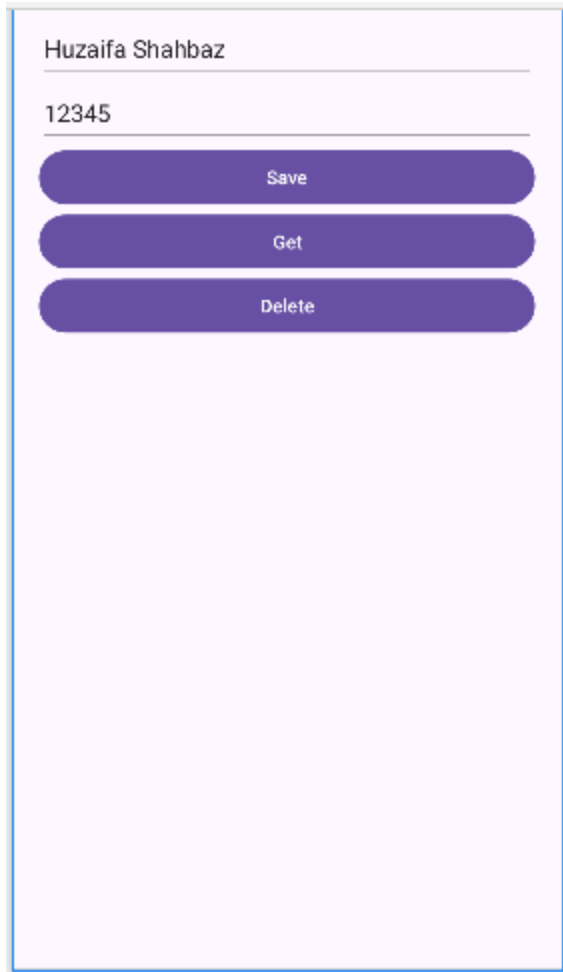
```

```
        SharedPreferences prefs = context.getSharedPreferences("MyPrefs",
            context.MODE_PRIVATE);
        SharedPreferences.Editor editor = prefs.edit();
        editor.remove(USER_NAME);
        editor.remove(PASS_WORD);
        return editor.commit();
    }
}
```

Results:



OUTPUT:



Home Task:

Use Tasbeeh app and use shared preference to store history.

Note: Shared Preference can only store strings.

JAVA CODE OF MAIN ACTIVITY:

```
package com.huzaifashahbaz.tasbeehappsharedpreferences;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.ColorUtils;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
```

```

import androidx.core.view.WindowInsetsCompat;
import java.util.ArrayList;
public class MainActivity extends AppCompatActivity {
    private TextView countView;//initailize the textview
    private Button countButton;//initialize the countbutton
    private Button resetButton;//initialize the resetbutton
    private Button viewHistoryButton;//initialize the viewhistoryButton
    private int count=0;//initialize count
    private ArrayList<CountEntry>countHistory;//initialize arraylist of counthistory
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);//main activity java file
        countHistory=CountSharedPreferences.getCountHistory(this);
        //DEFINE UI COMPONENTS
        countView=findViewById(R.id.count);//countview java file
        countButton=findViewById(R.id.count_btn);//count_btn java file
        resetButton=findViewById(R.id.reset_btn);//reset_btn java file
        viewHistoryButton=findViewById(R.id.history_btn);//history_btn java file
        //SET UP INITIAL COUNT
        updateCountView();
        //BUTTON LISTENERS
        //COUNTBUTTON CLICKLISTENER FUNCTION
        countButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                count++;
                updateCountView();
            }
        });
        //RESETBUTTON CLICKLISTENER FUNCTION
        resetButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                countHistory.add(new CountEntry(count, System.currentTimeMillis()));

CountSharedPreferences.saveCountHistory(MainActivity.this,countHistory);
                count=0;
                updateCountView();
            }
        });
    }
}

```



```

});
//HISTORYBUTTON ONCLICKLISTENERFUNCTION
viewHistoryButton.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        Intent countIntent=new Intent(MainActivity.this,HistoryActivity.class);
        startActivity(countIntent);
    }
});
}
private void updateCountView()
{
    countView.setText("Count:"+count);
}
}

```

JAVA CODE OF HISTORY ACTIVITY:

```

package com.huzaifashahbaz.tasbeehappsharedpreferences;
import android.content.DialogInterface;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ListView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import java.text.SimpleDateFormat;
import java.util.ArrayList;
import java.util.Date;
import java.util.Locale;

public class HistoryActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_history);//historyactivity javafile create
    }
}

```

```

        ListView historylist=findViewById(R.id.count_list);//count_list java file
create
        ArrayList <CountEntry> countHistory=
        CountSharedPreferences.getCountHistory(this);//arraylist
        if (CountHistory !=null)
        {
            CountHistoryAdapter historyAdapter = new
            CountHistoryAdapter(this,R.layout.count_list_item,countHistory);//counthistoryad
apter
            historylist.setAdapter(historyAdapter);
            historylist.setOnItemClickListener(new
            AdapterView.OnItemClickListener() {
                @Override
                public void onItemClick(AdapterView<?> adapterView, View view, int
                i, long l) {
                    showDateTimeDialog(countHistory.get(position));//datetimedialog
                }
            });
        }
        private void showDateTimeDialog(countEntry entry)
        {
            String formattedDate=formatDate(entry.getTimestamp());//string of formatted
date
            String dialogMessage="Count saved at:" + formattedDate;//string of
dialogMessage
            //build and show alerdialog
            AlertDialog.Builder builder=new AlertDialog.Builder(this);
            builder.setMessage(dialogMessage);
            builder.setMessage(dialogMessage)
                .setTitle("Count Information")//title of builder
                .setPositiveButton("OK", new DialogInterface.OnClickListener()
                {//positive button of dialog interface
                    public void onClick(DialogInterface dialogInterface, int i) {
                        //user clicked on button
                        dialog.dismiss();
                    }
                });
            AlertDialog dialog= builder.create();//buildercreate of dialog
            dialog.show();

```

```

    }
    private String formatDate(long timestamp)
    {
        SimpleDateFormat sdf=new SimpleDateFormat("yyyy-MM-dd HH:mm:ss",
Locale.getDefault());
        return sdf.format(new Date(timestamp));
    }
}

```

CountsharedPrefernces:

```

package com.huzaiifashahbaz.tasbeehappsharedpreferences;
import android.content.Context;
import android.content.SharedPreferences;

import java.util.ArrayList;
import java.util.HashSet;
import java.util.Set;

public class CountSharedPreferences {
    private static final String PREF_NAME="CountPreferences";//string prefname
    private static final String COUNT_HISTORY_KEY="countHistory";//string
counthistory
    public static void saveCountHistory(Context context, ArrayList<CountEntry>
countHistory)
    {
        SharedPreferences.Editor
editor=context.getSharedPreferences(PREF_NAME,context.MODE_PRIVATE).edi
t();//shared[reference editor
        Set<String> countHistorySet=new HashSet<>();//counthistoryset
        for (CountEntry entry:CountHistory)//for loop
        {
            countHistorySet.add(serializeCountEntry(entry));
        }
        editor.putString(COUNT_HISTORY_KEY,countHistorySet);
        editor.apply();
    }
    public static ArrayList<CountEntry>getCountHistory(Context context)
    {
        SharedPreferences prefs=
context.getSharedPreferences(PREF_NAME,Context.MODE_PRIVATE);
    }
}

```

```

        Set<String>
        countHistorySet=prefs.getStringSet(COUNT_HISTORY_KEY,null);
        if (countHistorySet !=null)
        {
            ArrayList<CountEntry> countHistory=new ArrayList<>();
            for (String serializedEntry:countHistory)
            {
                countHistory.add(deserializedCountEntry(serializedEntry));
            }
            return countHistory;
        }
        return new ArrayList<>();
    }
    private static String serializeCountEntry(CountEntry entry)
    {
        return entry.getCount()+ ","+entry.getTimestamp();
    }
    private static CountEntry deserializedCountEntry(String serializedEntry)
    {
        String[] parts =serializedEntry.split(",");
        int count=Integer.parseInt(parts[0]);
        long timestamp =Long.parseLong(parts[1]);
        return new CountEntry(count,timestamp);
    }
}

```

XML CODE OF MAIN ACTIVITY:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="2mm"
    tools:context=".MainActivity">
    <LinearLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginHorizontal="14mm"

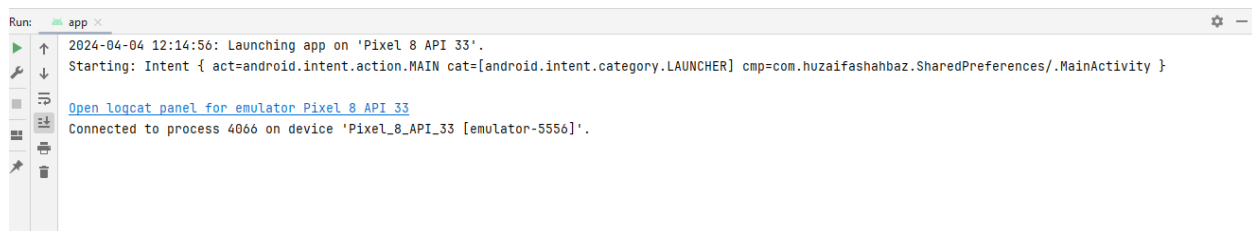
```

```
android:layout_marginTop="14mm"
android:orientation="vertical"
android:layout_gravity="center">
<TextView
    android:id="@+id/app_title"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:gravity="center"
    android:textSize="35sp"
    android:textStyle="bold"
    android:fontfamily="causal"
    android:text="Tasbeeh App"/>
<TextView
    android:id="@+id/count"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="15mm"
    android:gravity="center"
    android:textSize="25sp"
    android:textStyle="bold"
    android:fontFamily="monospace"
    android:text="Count:0"/>
<LinearLayout
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_gravity="center"
    android:layout_marginTop="14mm">
    <Button
        android:id="@+id/count_btn"
        android:layout_width="wrap_content"
        android:layout_height="match_parent"
        android:layout_marginHorizontal="1mm"
        android:text="Alhamdulillah"
        android:textSize="18sp"
        android:fontFamily="casual"
        android:textStyle="bold"/>
    <Button
        android:id="@+id/reset_btn"
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:layout_marginHorizontal="1mm"
        android:text="Reset"
        android:textSize="18sp"
        android:fontFamily="casual"
        android:textStyle="bold"/>
    </LinearLayout>
    <LinearLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:layout_gravity="center"
        android:layout_marginTop="14mm">
        <Button
            android:id="@+id/history_btn"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginHorizontal="18sp"
            android:fontFamily="casual"
            android:textStyle="bold"/>

    </LinearLayout>
</LinearLayout>
</LinearLayout>
```

Results:



OUTPUT:

Tasbeeh History

Count 1: 5 Times

Saved at: 2023-12-04 22:36:27

Count 2: 9 Times

Saved at: 2023-12-04 22:43:44

Count 3: 6 Times

Saved at: 2023-12-04 22:36:26

Count 4: 33 Times

Saved at: 2023-12-04 22:36:48

Conclusion:

In conclusion, we had learned the shared preferences in mobile app. Also we know the working and function of tasbeehcounter and student form by using Java Code and XML linear layout.